

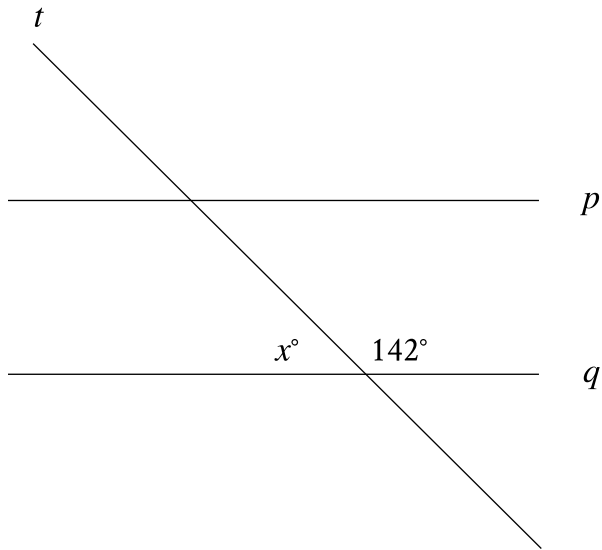
● EASY

● GEOMETRY AND TRIGONOMETRY

Geometry and Trigonometry

06

10 questions · ~15 min



Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled t , p , and q .
- Line t intersects both line p and line q .
- At the intersection of line t and line q , 2 angles are labeled clockwise from top left as follows:
 - Top left: x°
 - Top right: 142°
- A note indicates the figure is not drawn to scale.

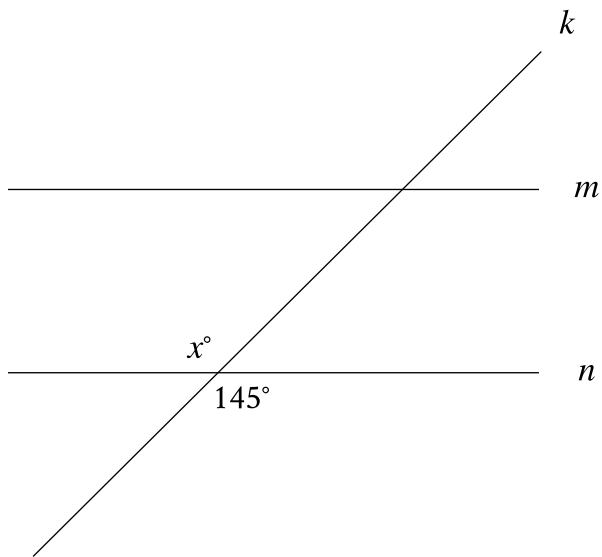
In the figure, line p is parallel to line q , and line t intersects both lines. What is the value of $x + 142$?

- A. 52
- B. 90
- C. 142

D. 180

What is the area of a rectangle with a length of 17 centimeters cm and a width of 7 cm?

- A. 24 cm^2
- B. 48 cm^2
- C. 119 cm^2
- D. 576 cm^2



Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled k , m , and n .
- Line k intersects both line m and line n .
- At the intersection of line k and line n , 2 angles are labeled clockwise from top left as follows:
 - Top left: x°
 - Bottom right: 145°
- A note indicates the figure is not drawn to scale.

In the figure, line m is parallel to line n , and line k intersects both lines. Which of the following statements is true?

- A. The value of x is less than 145.
- B. The value of x is greater than 145.
- C. The value of x is equal to 145.

D. The value of x cannot be determined.

Q4

GEOMETRY AND TRIGONOMETRY

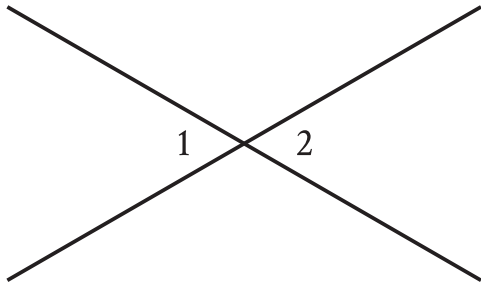
Triangle R has an area of 80 square centimeters cm^2 . Square S has side lengths of 4 cm. What is the total area of triangle R and square S, in cm^2 ?

A. 42

B. 44

C. 84

D. 96

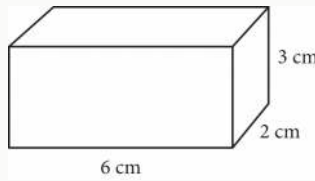


Note: Figure not drawn to scale.

- At the intersection of the 2 lines, the angles are labeled clockwise from left as follows:
 - Left: 1
 - Top: unlabeled
 - Right: 2
 - Bottom: unlabeled
- A note indicates the figure is not drawn to scale.

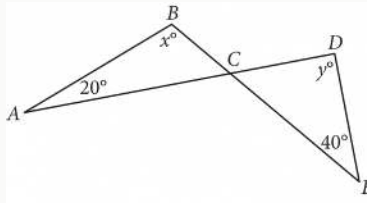
In the figure, two lines intersect at a point. Angle 1 and angle 2 are vertical angles. The measure of angle 1 is 72° . What is the measure of angle 2?

- A. 72°
- B. 108°
- C. 144°
- D. 288°



The figure shows the lengths, in centimeters (cm), of the edges of a right rectangular prism. The volume V of a right rectangular prism is ℓwh , where ℓ is the length of the prism, w is the width of the prism, and h is the height of the prism. What is the volume, in cubic centimeters, of the prism?

- A. 36
- B. 24
- C. 12
- D. 11



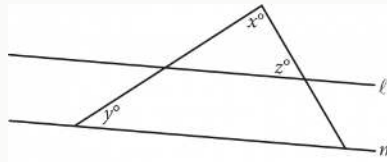
Note: Figure not drawn to scale.

In the figure above, $\overline{AD}$ intersects $\overline{BE}$ at C. If $x = 100$, what is the value of y ?

- A. 100
- B. 90
- C. 80
- D. 60

Triangle ABC and triangle DEF are similar triangles, where $\overline{AB}$ and $\overline{DE}$ are corresponding sides. If $\overline{DE} = 2\overline{AB}$ and the perimeter of triangle ABC is 20, what is the perimeter of triangle DEF ?

- A. 10
- B. 40
- C. 80
- D. 120



Note: Figure not drawn to scale.

In the figure above, lines l and m are parallel, $y = 20$, and $z = 60$. What is the value of x ?

- A. 120
- B. 100
- C. 90
- D. 80

Q10

GRID-IN

GEOMETRY AND TRIGONOMETRY

The area of a rectangle is 57 square inches. The length of the longest side of the rectangle is 19 inches. What is the length, in inches, of the shortest side of this rectangle?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.